Fissure Gas Samples, April 2026 – Notes for Phillip

Steamboat Hills Geothermal Outflow Monitoring – UNR Thesis Project

2026-10-15

Background

Steamboat Hills has a fairly long geochemistry record for a Great Basin system – White and coworkers characterized it geologically and hydrologically starting in the 1950s-60s, and there's a run of gas and water chemistry papers from the 1970s-90s: Nehring (1979, 1980) on reservoir temperature and gas composition, Mariner and Janik (1995) with water and gas analyses for several CPI production wells (including well 23-5, which ends up being our main point of comparison below), and Arehart, Coolbaugh, and Poulson (2003) arguing for a magmatic heat source using trace elements and gas geochemistry. I've been building a database that pulls all of this together along with NDEP, NDWR, and our own field data, mostly for the chloride-discharge side of the thesis – but it also means we have a real, if sparse, historical gas-chemistry baseline to check new samples against.

The 2025 eruption and the fissure

In June 2025 the Lower Sinter Terrace had a geyser-like eruption that opened a new fissure (Lindsey et al. 2026 give the fuller account of how the activity built up over 2022-2025 before that). The well was capped not long after, but discharge didn't stop – it moved laterally along the fracture system instead, and the fissure has kept degassing/discharging since. Cary and I put together a poster on this for GRC (Irvin and Lindsey, 2026) arguing, mostly from paired temperature loggers and a Cl mass-balance estimate, that this looks more like redistribution within an existing shallow conduit system than a brand-new source turning on.

Sampling

You, Tobias, and Karim collected three gas samples at and around the fissure on April 29-30, 2026 – two points right at the fissure and a third across the road at a separate degassing spot.

Sample	Date	Approx. location	Field temp (C)	Notes
NV26-1	Apr 29	at the fissure	99.5	co-located with a water sample we collected May 1 (see below)
NV26-2	Apr 30	fissure, second point	96	flagged – possible bottle leak noted in the field, and this is also where sampling went through fresh asphalt that the gas was escaping through
NV26-3	Apr 30	across the road	96	second degassing area

NV26-2 is the one I want your read on. We're not throwing it out, but we're also not treating it the same as the other two until we hear back on whether the leak and/or the asphalt breakthrough would meaningfully change the result.

What we're seeing so far

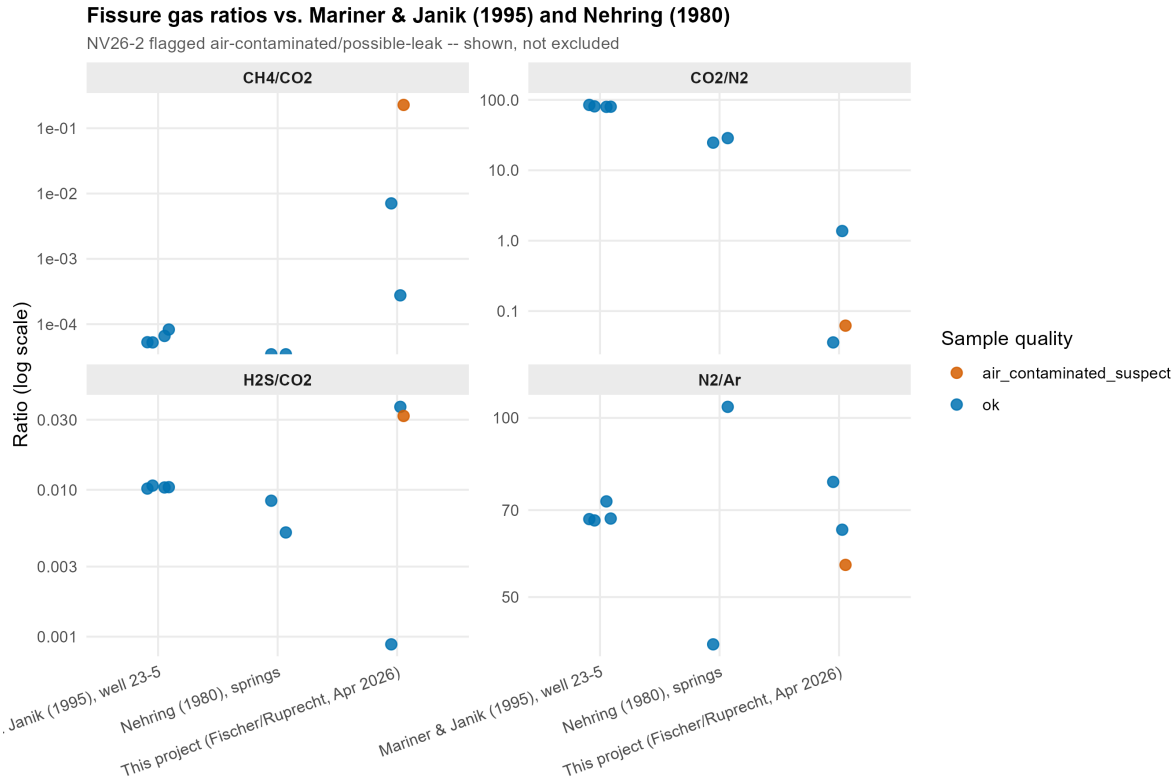
This is exploratory – these are three samples, one of them flagged, so I'm treating all of it as a first look rather than a result to lean on hard.

Air correction and geothermometry. Surface gas is never air-free, so before doing anything else we corrected for atmospheric contamination (standard approach: all measured O2 treated as air, air-derived N2 subtracted using the dry-air N2/O2 ratio, everything renormalized). All three samples turn out to have a real amount of air mixed in – not surprising for

shallow vent sampling instead of a wellhead – but NV26-2 is essentially explainable by air alone.

Sample	Air fraction (O2-based)	D'Amore-Panichi T (C)
NV26-1	53%	228
NV26-2	100%	209 (not trusted – see above)
NV26-3	64%	181

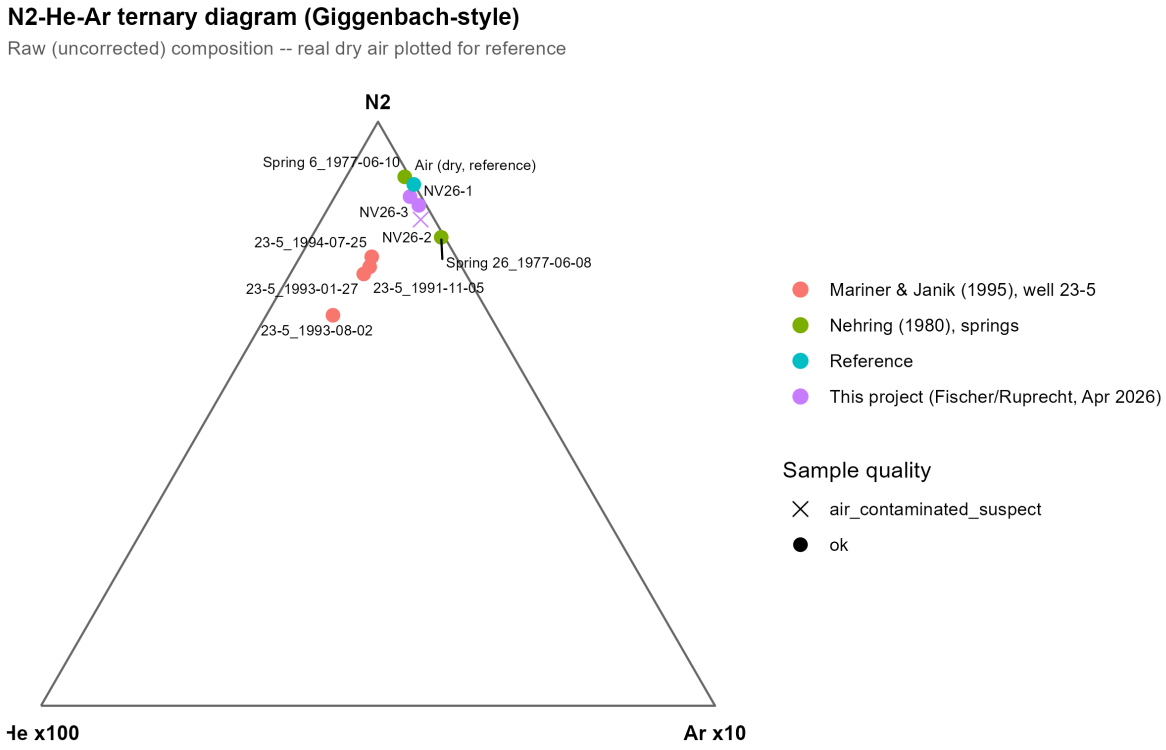
For reference, Mariner and Janik's well 23-5 T-DP values from the 1990s run 173-200 C, so NV26-1 and NV26-3 bracket that range – and both are well above the measured field temperatures (96-99.5 C), which is the same pattern we're already seeing in the water geothermometers (quartz below measured temperature, Na/K above it). It's a reasonable result, but three samples isn't much to build on yet.



How the two "good" samples compare to Nehring and to Mariner & Janik. CO2/N2, H2S/CO2, N2/Ar, and CH4/CO2 for NV26-1 and NV26-3 land closer to Nehring's (1980) spring samples than to well 23-5 – which makes sense, since Nehring sampled surface springs too, not a wellhead, so a similar degree of air mixing isn't surprising. NV26-2 sits off on its own in every panel, again consistent with it being mostly air. Nehring and Mariner &

Janik only overlap with us on major-gas composition – neither reports a helium isotope ratio, so that comparison stops at CO2/H2S/N2/Ar/CH4.

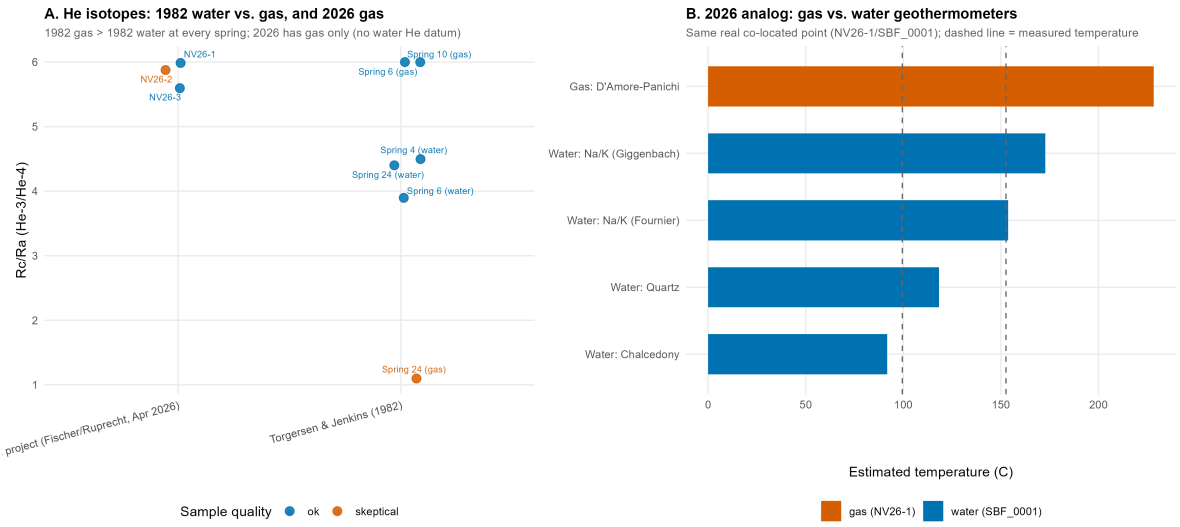
A ternary view of the same thing. The classic way to look at this kind of air-mixing question is an N2-He-Ar diagram (the same style Mariner & Janik use in their own Figure 9). I plotted our three points against well 23-5, the two usable Nehring rows, and real dry air for reference.



All three of our points, and Nehring's two spring samples, cluster right next to the air point – well 23-5 sits noticeably further away, toward more N2 relative to Ar and He. That lines up with the air correction above: surface vent/spring gas looks a lot more air-like than gas collected at a wellhead, which isn't a new idea but it's nice to see it fall out of the data this cleanly. I used the common He-times-100/Ar-times-10 scaling to make the plot readable, but I haven't gone back to a primary Giggenbach reference to confirm that's the right scaling or

where the "real" air/crustal/mantle field boundaries are supposed to sit on a plot like this – flagged as a question below rather than something I'm presenting as settled.

Helium isotopes – it turns out there's a real 1982 baseline, and it's a surprisingly tight match. I'd first written this saying nobody had ever published a He-3/He-4 ratio for Steamboat – wrong, and I'm glad you (or whoever flagged it) caught it, because the real source turned out to be even better than what I'd patched it with the first time. Torgersen and Jenkins (1982, GCA) measured He isotopes in both water and gas at four Steamboat springs. Their two gas samples they themselves call the more reliable ones (Springs 6 and 10) both come out at Rc/Ra = 6.0 – our three new samples come out at 5.6-6.0. That's about as close a match as you could ask for across 44 years, and it's a gas-to-gas comparison specifically, which matters since their own water-phase numbers run lower (3.9-4.5) and two of their gas values are flagged in their own text as suspect (air mixing or the vent contaminating itself). See the figure below. Kennedy & van Soest (2007, Science), which is what I'd originally found this through (via Dhakal et al.'s citation of it, not read directly at first), also names Steamboat as a high-R/Ra Basin and Range site, but that's a regional paper, not Steamboat-specific – Torgersen and Jenkins is the real, direct comparison. Arehart et al. (2003) make basically the same magmatic-source argument from trace elements and gas geochemistry, but cite a different, earlier Kennedy paper (2002, not the 2007 one) and don't give a gas composition table or a He number of their own, so they're interpretive context here rather than a number to compare against.

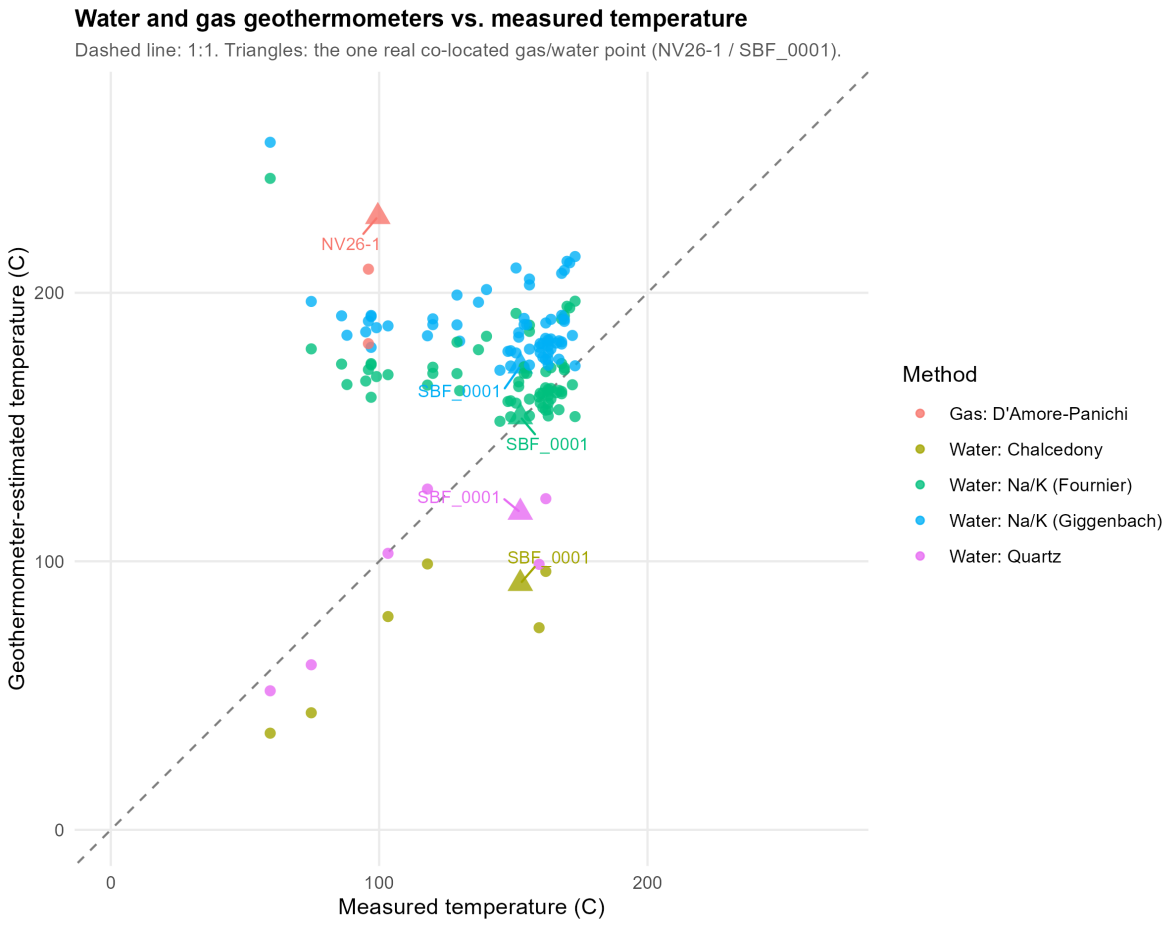


Panel A is the He-isotope comparison above. Panel B is the closest thing I can show for 2026, since we don't have a dissolved-gas/He measurement on the water side this time – it's the gas vs. water temperature geothermometer comparison at this same NV26-1/SBF_0001 point instead. The pattern rhymes: in 1982 every spring's gas phase reads a higher Rc/Ra than its own water phase, and in 2026 the gas geothermometer reads hotter than every water method except Na/K. Different measurement (isotope ratio vs. temperature), same qualitative story –

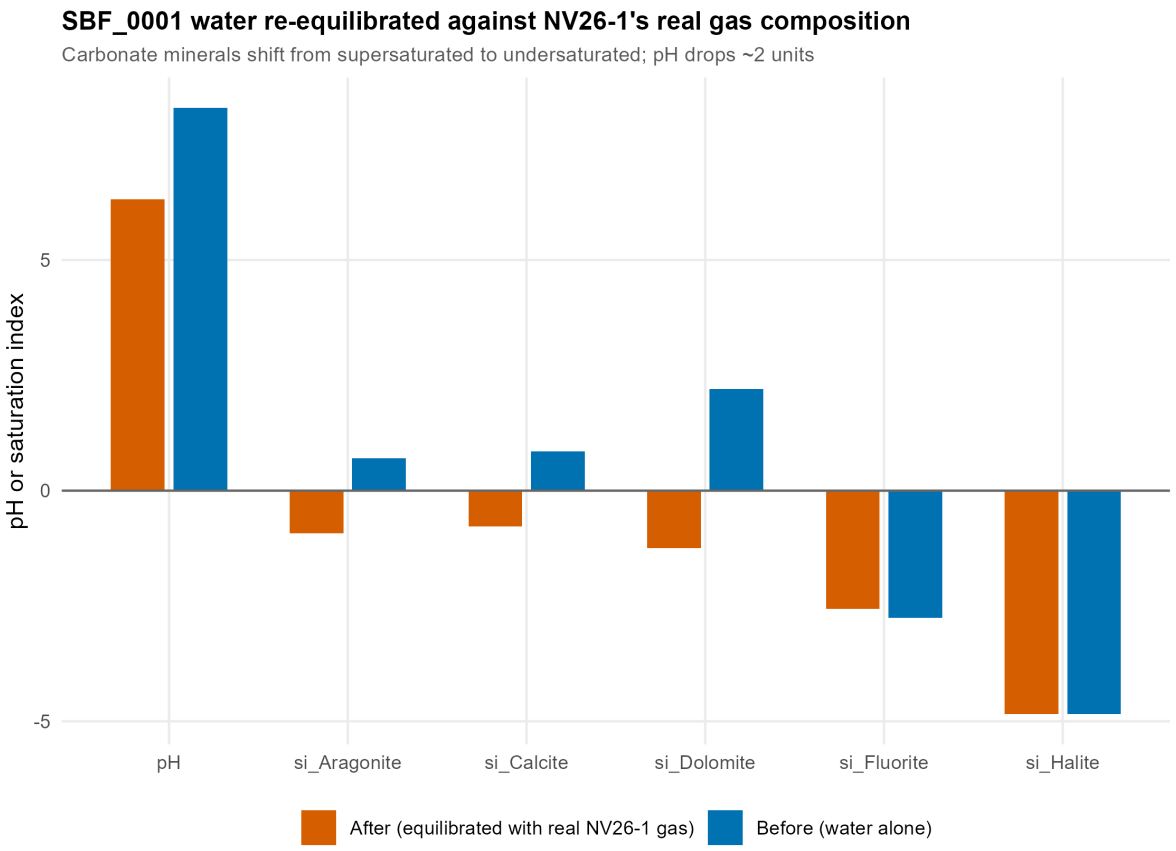
the gas phase is consistently telling you something different from, and arguably deeper/hotter than, the water at the same point, in both eras.

One more thing worth a mention. NV26-1 sits right on top of SBF_0001, a water sample we collected two days later (May 1) with full major-ion and isotope chemistry – a mature Na-Cl thermal water (Cl 815 mg/L). It's the first spot in the project with both phases from essentially the same location, which matters more for the mixing modeling described below than for anything in this section.

Water vs. gas geothermometry, and a first real PHREEQC run. Putting the gas geothermometer estimate next to our existing water geothermometers (Na/K, quartz, chalcedony – 148 estimates across 67 real thermal samples) for the one point where we have both, NV26-1's gas estimate (228 C) lands closest to the Na/K water estimates (153-173 C), not the quartz/chalcedony ones (92-118 C) – both the gas and the Na/K signature seem to be reading a similar, deeper part of the system than silica does.



I also went ahead and ran a real PHREEQC `GAS_PHASE` equilibration – took SBF_0001's water chemistry and re-equilibrated it against a gas phase seeded with NV26-1's real, air-corrected CO2:H2S ratio (96.5:3.5), at 1 atmosphere, treated as an open system (i.e. assuming the fissure keeps resupplying gas rather than it being a small fixed pocket).



The result: pH drops from the water's measured 8.3 down to a modeled 6.3, and calcite/aragonite/dolomite all flip from supersaturated to undersaturated. My read is that this means the water as sampled isn't actually sitting in equilibrium with the gas as sampled – which could mean they're not really the same fluid parcel, that gas and liquid separate out before either one reaches the surface, or that the real reservoir pressure (not the 1 atm I used, which was a simplification) changes things. I don't have the background to know which of those is most likely, which is part of why I'm asking below.

How this fits into the rest of the thesis

The short version is that this is the first real gas dataset the project has, and it slots into a few places we'd already scoped out but had no real data for:

- The D'Amore-Panichi geothermometer work above reuses code that had only ever been run against your group's own historical numbers (Mariner & Janik) as a check – this is the first time it's been run on a Steamboat sample we actually collected.

- We'd had a PHREEQC gas-phase/mixing setup built for a while, waiting on a real dissolved-gas sample to test it against – NV26-1 is the first one, and the real run above is already a result, not just a capability anymore.
- The NV26-1/SBF_0001 pairing is a natural input to the mixing-model chapter – a real gas end-member next to a real water end-member from (almost) the same spot and the same week.
- More broadly, it's a second, independent way to ask whether the system has cooled/diluted the way the water geochemistry suggests, which matters for the "redistribution vs. new source" question the poster raised.

Questions for you

Mostly I want to make sure we're using and describing your group's work correctly before any of this goes further:

- How should we cite the gas analyses in the thesis, and separately if this turns into a conference abstract or paper – Fischer's lab and Barry's lab specifically, or is there a preferred citation you'd want us to use?
- Is the air correction above a reasonable way to handle Giggenbach bottle samples collected this way, or does your group normally do something different that we should match?
- On the PHREEQC gas-phase run – is 1 atm/open-system a reasonable starting assumption for this, or is there a better estimate of the real pressure/volume this gas phase should represent at depth? And is the disequilibrium result (pH/carbonate SI shifting a lot) actually surprising, or is that just what you'd expect for a water and gas sampled two days and some distance apart rather than from the exact same parcel?
- On the ternary plot – is the He-times-100/Ar-times-10 scaling I used the right convention, and is there a reference version of that diagram (with real air/crustal/mantle field boundaries) I should be plotting these points against instead of just showing them relative to a single air point?
- On NV26-2 – is the leak serious enough that the sample shouldn't be used at all, or is it still usable with the caveat we've got on it?
- Should we hold off on presenting any interpretation of the noble gas / carbon isotope numbers until you, Tobias, or Karim have had a chance to look at them, or is what's here reasonable to go forward with as a preliminary read?
- Would you, Tobias, and/or Karim want to be coauthors if this turns into a conference abstract or paper down the line?

Contact

Tyler Irvin, UNR – happy to send the underlying data/derived CSVs if useful on your end.